

Textual Analysis in Finance - Final project

Text Data and Stock Market Impact

May 5, 2026

1 Overview

The final project examines how textual features extracted from financial documents influence stock market behavior. In addition to demonstrating technical proficiency, the project should include a critical assessment of the broader implications of using financial text data in market analysis.

Students are required to select and complete ONLY ONE of the following project options; i.e., Section 2.1 or Section 2.2.

2 Projects' description

2.1 Central bank communication and market impact

This project investigates how Federal Reserve communications influence market volatility. By applying textual analysis techniques to official FOMC press releases from 2023 to 2025, the study aims to examine the relationship between the sentiment/topics of these communications and subsequent market reactions.

2.1.1 Data description

This project involves two datasets: **FOMC press releases** and **the intraday US total market index**.

- **FOMC press releases:** The data can be downloaded [here](#). There are **24 events** between 2023 and 2025. For each FOMC meeting,
 - Under the **Statement** section, download the PDF file, which is the press releases
 - In the statement, you can also see the exact time it is released. This timestamp is important to study the intraday reactions of the stock market.
- **The intraday US total market index:** The second-by-second US total market index from 09:30:00 01 January 2023 to 16:00:00 31 December 2025. The data is provided [here](#).

2.1.2 Textual analysis

The textual analysis must comprise the following steps:

- **Preprocessing:**

- After downloading all press releases, organize the data in the following format:

ID	Text	Time
1	Recent indicators suggest that economic activity has continued to expand at a solid pace	20250319 14:00:00
...	...	...

- Normalize the data

- **Topic and sentiment extraction:**

- Extract topics discussed in those releases.
- Measure sentiment, i.e., dovish, hawkish, or neutral, of those releases using dictionary-based approaches/language models/large language models.

2.1.3 Questions

The analysis should address the following core questions:

- Does the market systematically react to FOMC press releases?
- Is the market response influenced by the sentiment/topic expressed in FOMC press releases?

Beyond these core questions, students are encouraged to extend their analysis by formulating additional research questions and exploring further insights that may emerge from the data.

2.1.4 References

Anderson, M. (2010). Using intraday data to gauge financial market response to Federal Reserve and ECB monetary policy decisions. *International Journal of Central Banking*, 6(2), 107-116. [[Article link](#)].

2.2 Open topic

Students may choose a topic different from the one outlined above for the final project. However, all submissions must satisfy the overarching requirement: **the project should demonstrate how financial text data informs/influences financial outcomes**. In particular, students are expected to apply appropriate text analysis techniques and clearly illustrate how text-based information relates to financial decisions or outcomes. e.g., stock markets, corporate governance, corporate investment decisions, etc.

Students are welcome to build on work from other courses or their ongoing thesis. Ultimately, the course aims to equip students with the skills to effectively utilize textual data in financial research and decision-making.

For students who choose an open topic, the report must include the following components:

- **Motivation and Research Questions:**
 - Clearly state the primary research questions
 - Explain why textual data is appropriate for addressing these questions
- **Data Description:** Describe the data collection process, including the sources of the data and how it was obtained
- **Text Analysis Techniques:**
 - Specify the types of information to be extracted from the text data
 - Explain the methods and techniques used to extract this information
- **Link to Financial Outcomes:** Analyze and present the relationship between the extracted textual information and the relevant financial decisions or outcomes, and use this analysis to answer the primary research questions.

3 Deliverables and deadline

The final submission should consist of:

- **Code script:** Well-documented scripts detailing the data processing, analysis, regression, and visualization.
- **Analysis report:** A comprehensive report that explains the methodology, analysis, results, and conclusions.

- ONLY FOR STUDENTS CHOOSING THE OPEN TOPIC: Besides the code and report, your data must be provided.
 - If the data is $\leq 5\text{MB}$, attach the data to the submission
 - If the data is $> 5\text{MB}$, upload it to Google Drive or Dropbox, then provide the link to download the data in the code script

THE DATA IS USED FOR GRADING PURPOSES ONLY.

Submission information:

- **Deadline: Sunday, June 7, 2025, at 23:59 CEST**
- Submit by email to: triminh.phan@unibas.ch
- Email subject:
[Textual Analysis in Finance] Final project - <Surname_1>_<Surname_2>_<Surname_3>

4 Requirements and evaluation criteria

4.1 Submission requirements

The submission must meet the following requirements:

- The report must include a title page containing:
 - The project title;
 - The full names (*Surname, First Name*), University of Basel email addresses, and matriculation numbers of all group members
- The report may include up to **three** tables and **three** figures. All tables or figures should be placed in the appendices and properly referenced in the main text.
- The main text must not exceed 5 A4 pages, using font size 12pt and one-and-a-half line spacing. This page limit excludes the title page and any appendices.

4.2 Evaluation

- **Reproducibility:** All analyses and results presented in the report **must be fully reproducible**.
- **Analytical Rigor:** Comprehensive and logical data preprocessing; effective application of textual analysis and statistical techniques.
- **Methodological Clarity:** Clear, well-documented code . The code must demonstrate how the data are loaded and processed, and how all reported outputs (e.g., figures, tables, and regression results) are generated from the raw data. A .txt file instructing how to run the code is highly recommended.
- **Report Quality:** A logically structured analysis report linking textual attributes to financial outcomes. The report should convincingly state the problems and the relevance of the text data.